line between the A/C condenser and the A/C evaporator. This restriction established the pressure differential between the high and low-pressure sides of the A/C system.

The inlet end of the fixed A/C orifice tube has a nylon mesh filter screen, which filters the refrigerant and helps to reduce the potential for blockage of the metering orifice by refrigerant system contaminants (Fig. 13). The outlet end of the tube has a nylon mesh diffuser screen. The O-rings on the plastic body of the tube seal it to the inside of the liquid line and prevent the refrigerant from bypassing the fixed metering orifice.

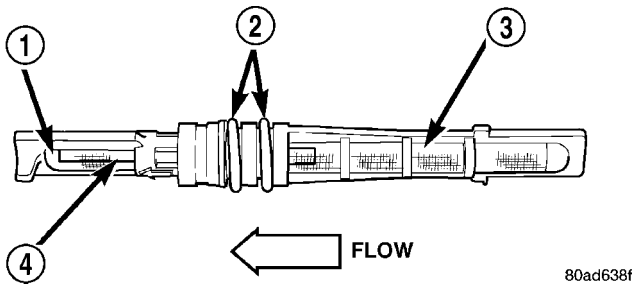

Fig. 13 Fixed A/C Orifice Tube - Typical

1 - DIFFUSER SCREEN
2 - O-RING SEALS
3 - INLET FILTER SCREEN
4 - ORIFICE

OPERATION

The fixed A/C orifice tube is used to meter the flow of liquid refrigerant into the A/C evaporator. The high-pressure liquid refrigerant from the A/C condenser expands into a low-pressure liquid as it passes through the metering orifice and diffuser screen of the A/C orifice tube.

The A/C orifice tube is not serviceable. It cannot be repaired, and if faulty or plugged, it must be replaced as part of the A/C liquid line (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING/LIQUID LINE - REMOVAL).

DIAGNOSIS AND TESTING

A / C ORIFICE TUBE

WARNING: The A/C liquid line between the A/C condenser and the A/C orifice tube can become hot enough to burn the skin. Use extreme caution when performing the following test to prevent possible personal injury.

NOTE: The A/C orifice tube can be checked for proper operation using the following procedure. However, the A/C orifice tube is only serviced as a part of the A/C liquid line. If the results of this test indicate that the A/C orifice tube is obstructed or missing, the A/C liquid line must be replaced.

(1) Confirm that the refrigerant system is properly charged (Refer to 24 - HEATING & AIR CONDITIONING - DIAGNOSIS AND TESTING - A/C PERFORMANCE TEST).

(2) Start the engine. Turn on the A/C system and confirm that the compressor clutch is engaged.

(3) Allow the A/C system to operate for five minutes.

(4) Lightly and cautiously touch the A/C liquid line near the condenser outlet at the front of the engine compartment. The liquid line should be hot to the touch.

(5) Touch the A/C liquid line near the evaporator inlet at the rear of the engine compartment. The liquid line should be cold to the touch.

(6) If there is a distinct temperature differential between the two ends of the A/C liquid line, the A/C orifice tube is in good condition. If there is little or no detectable temperature differential between the two ends of the A/C liquid line, the A/C orifice tube is obstructed or missing and the A/C liquid line must be replaced (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING/A/C LIQUID LINE - REMOVAL).

ACCUMULATOR

DESCRIPTION

The A/C accumulator (Fig. 14) is mounted in the engine compartment between the A/C evaporator and the A/C compressor. An integral mounting bracket is used to secure the A/C accumulator to the frame rail. The A/C low pressure switch is mounted on a fitting located on the top of the A/C accumulator.

The A/C accumulator cannot be repaired and, if faulty or damaged, it must be replaced. The rubber O-rings used on accumulator connections and the A/C low pressure switch are available for service replacement.